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Experiment 3:

Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

ID3 Algorithm:

Code:

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
df = pd.read_csv("training_1.csv")
le = LabelEncoder()
for col in df.columns:
    df[col] = le.fit_transform(df[col])
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
model = DecisionTreeClassifier(criterion="entropy")
model.fit(X, y)
sample = pd.DataFrame([[2, 0, 1, 1, 2, 0, 1]], columns=X.columns)
result = model.predict(sample)
if result[0] == 1:
    print("Play Tennis = Yes")
else:
    print("Play Tennis = No")
```

Output:

```
>> | ===== RESTART: C:\Users\
    | Play Tennis = Yes
>> |
```